Analysis Phase: Building a Blog with Vite + React, Strapi, and Tailwind CSS

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Abstract

This analysis includes gathering user stories and opinions, monitoring existing products and solutions, measuring current systems potential capacity limitations, and evaluating potential business implications to define various end-users and stakeholder requirements in building a successful development blog platform that scales past its current planned iterations. We found that the current systems will support the requirements mentioned, but also acknowledged that if stakeholder interest grows and expands, a larger scale deployment to a VPS system would be needed to handle future bandwidth limitations, stakeholder requirements, and business implications. We addressed how each of the technologies used in our stack, VRTS, will aid in delivering a system that simply, securely, and efficiently meets users professional needs!

Analysis Phase: Building a Blog with Vite + React, Strapi, and Tailwind CSS

Utilising a modern tech stack - Vite+React, TailwindCSS, and Strapi, this project aims to develop a blog platform that can be easily maintained over time and serve as extra documentation to users' projects and portfolios as they learn and continue to grow; additionally, serving as a dynamic and interactive showcase of the skills and expertise I've learned throughout my computer science program. The future target audience for whom may be developers, students, or professionals looking for an interactive and visually appealing way to document their work and learning, but will be served locally with smaller ambitions of 2 home users for the time being, until later sprints are iterated to for a larger VPS deployment. This analysis serves to define the various stakeholders who may be interested in this project; as well as it serves to outline the various features/functions and non-functional/technical requirements needed to achieve said demand; as well as an analysis of the potential business implications such a product could entail!

The current project is only slated to serve two user profiles to a small subset of colleagues, friends and family, and potential employers; but will be set up locally in a way that is easily deployable to a larger VPS, or Virtual Private Server, in the future! This means the current potential daily load on the system may be small, maybe a max of 100 requests for the day; but as my skills, hopefully money, and community grow, I may want to move the system to a larger VPS system. I could do this to grow the community I hope to serve, empower other creators to be hosted or serve themselves, and potentially help people get employed or even make a potential business venture off the project! By building our site with Vite+React, TailwindCSS, Strapi, and NginX Proxy Manager on a linux container, we ensure the project is easily replicated to a larger VPS system and scales well to future stakeholders! The following analysis will define

end users, collaborative IT specialists, and business requirements in relation to current and potential future stakeholders respectively.

Analysis Section

Again, while the end users are currently a small subset of people, previously mentioned, the future stakeholders may include a growing community of collaborators and professionals, employers, or may even serve as a platform in of itself to a larger audience of collaborative professionals outside the current scope of this project. Their goals may include being able to document their works and studies through a blog-style format using clear controls, inputs, formats, text, images, embedded video, and code snippets! Strapi's locally hosted back-end will luckily handle all of this for us, to be later detailed in this document. These needs greatly shaped the decision to use the chosen tech stack, Vite + React, Strapi, and Tailwind CSS, which we will now refer to as VRTS. The site needs to be self-contained for future deployments, so we chose to use an Ubuntu linux container, on Proxmox, as our host for this project. This ensures an easy and compatible deployment to a larger VPS down the line, since the same technologies can be used there. By using Vite+React, we ensure a speedy and reactive deployment to the potential users browsers! By using TailwindCSS, we ensure a clear design philosophy to ensure clear reactive inputs to users, and with Strapi we ensure clear reactive controls to content creators and ensure scalability and security in that it is open-source and is frequently updated.

The IT specialists and collaborators involved in collecting said data included myself, as well as friends and colleagues who have also expressed interest in being able to more affordably and efficiently host their works and interest; and have provided me with their user-stories to build a set of functional/non-functional requirements for a platform that serves their needs and portfolios!

As part of the analysis phase, I reflected on the previous iteration by researching Strapi's API calls and setting up the environment with Vite on my local server within an instance of Visual Studio Code on an Ubuntu Linux container on Proxmox within my own locally hosted server; then, I spent the next few days talking to friends, colleagues, and other professionals to see what they've been struggling to show off, either due to limitations in their own time and budgets. Concisely, here's a few that stood out:

"I'd like to be able to just drop everything I'm working on for a particular post or topic into one thing, like code-snippets and various file types together!"

"I wanna be able to feature whatever I want at the top!"

"LinkedIn is great, but it's hard to format things how you want sometimes, especially if you aren't paying for their LinkedIn premium. Sites like Squarespace costs a bit much for just serving my portfolio and other solutions like Pinterest, Wordpress, and they either lack the features I want, or the costs."

- (Sanderson & Superior)

"I'd like to be able to just hop on and write, similar to 'archive of your own', but maybe with some more controls?"

- (Sanderson & Partner)

"Yeah, I'd just want to be able to show off all my certificates and portfolio[…]"

- (Sanderson & Family Member)

Nextly, some time was spent monitoring current solutions that support developer and general portfolios to see what was successful already. From the content management side of this project, Strapi handles the bulk of users requests, and then some, and scales per their open-source license agreement.

Convenient editor

Adding content to Strapi is clear and intuitive, you'll become a pro user in 2 minutes.

Component library

Use and reuse components to build custom pages from scratch. Learn more

Roles and permissions

Share access to the CMS with other team members and control what content they can read, edit, update or delete. Learn more

Media files support

Images, videos, audio files, documents - everything is stored in the media library. Learn more

Multilingual support

Create different versions of your websites or apps in different languages. Learn more

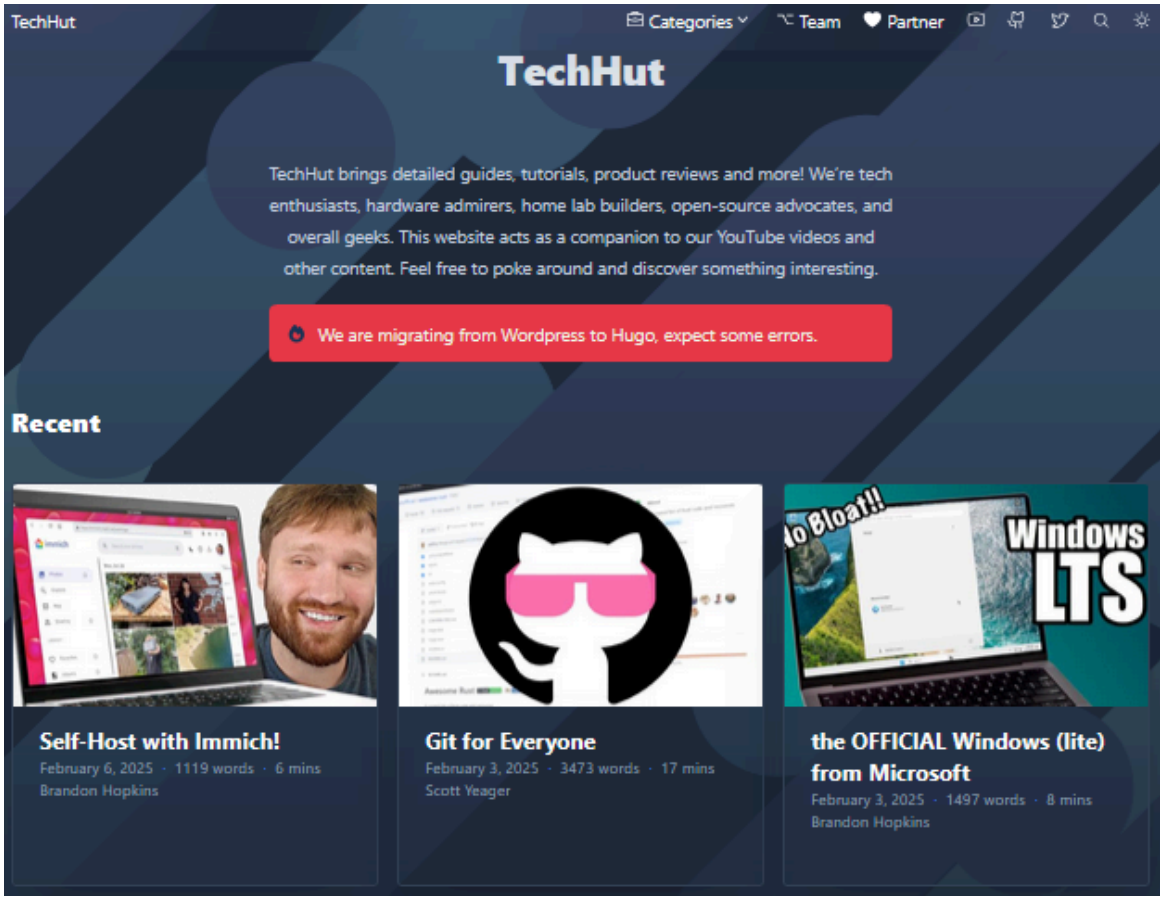
Omnichannel

Use Strapi as a single-source-of-truth for all of your websites, apps, or IoT projects.

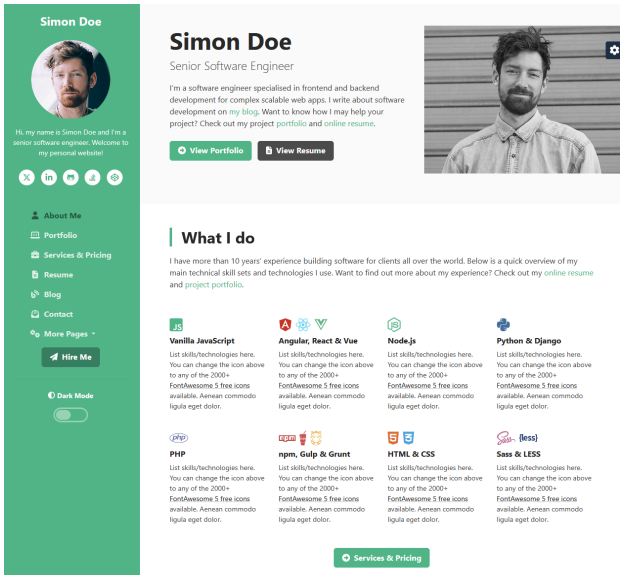
(*Strapi CMS for content teams*)

Without going into too much further detail at this time in the analysis phase, Vite+React and TailwindCSS will handle delivering said content to end-users, viewers and collaborators devices, quickly and seamlessly.

Furthermore, I monitored an array of sources that seemed to match users requirements in terms of the front-facing designs that can be implemented within our chosen tech-stack, VRTS. We found Techhut.tv's developer blog and various other blog templates had similar requirements to our own and compiled similarities between them based on some of the user stories we gained.



(TechHut)



Features

- Minimal styling
- Mobile responsive
- 100/100 Lighthouse performance
- SEO-friendly with canonical URLs and OpenGraph data
- Sitemap support
- RSS Feed support
- Markdown & MDX support
- Syntax highlighting
- Image optimization
- Table of contents
- Dark mode
- Reading Time
- Pagefind static search library integration
- Related posts
- Share posts (Linkedin, twitter)
- Draft mode
- Copy code block

(3rd Wave Media, 2024) (Daniel, 2023)

I conducted an analysis of what our current system can handle as well as an analysis of what a future system might require. Typically, VPS packages that support small-medium websites will provide 1-2 core cpus, 1-4gb of ram for the system, up to 500gb of storage space, and generally at least 500gb upload speed from their locations (Austin, 2025). My current configuration has 5 cores, 32gb ram, two 1tb solid state drives, two 16tb drives, and is currently underutilised (James, 2024), but I can only utilize 20mbs upload and am subject to Internet Service Provider contracts and fees should the project scale locally. For a typical blog, 7400 visitors per month could be supported on 2GB of bandwidth (C., 2022), but as I only plan to host maybe 30 per month, that bandwidth scales down to only 13-20 mb per month. At any one time I may be hit with maybe 15 users, based on my class size. A general blog page after compression may be up to 112KB-300kb, so 15 x 300KB = 4.5MB (Mirchevski), so even then my 20MB upload seems to cover this requirement for the time being.

The following specifies the Functional/Non-Functional requirements consolidated by myself, the IT specialist, based off said research:

Functional Requirements		**Non-Functional Requirements**	
<u>Blog</u>		**<u>Capacity</u>**	
The website shall display an About-Me Section.	Must Have	The website shall support a peak of 50 or less concurrent sessions at launch.	MustHave
The website shall display a Blog Section with recent posts.	Must Have	**<u>Availability</u>**	
The website shall display Social Media Sharing buttons.	Must Have	The system should be available 24 hrs per day, 334 days a year.	MustHave
The website shall display Portfolio/Project Links.	Must Have	The system shall not lose any user content.	MustHave
The website shall be sortable by Categories/Tags.	MustHave	The system shall reflect user changes within 5 minutes of	MustHave

		submission.	
The website must allow Embedded Code.	MustHave	The website should log users in within 5 seconds.	MustHave
The website shall have a Search Function.	MustHave		
The website should have a Comments Section per blog post.	ShouldHave		
The website should feature Responsive Design to accommodate a variety of devices. MustHave.	MustHave		
The website must feature SEO Optimization	MustHave		

These requirements seem to entail we'll have our main greeting page with the listed requirements above, as well as a blog page for each listed blog post or additional page. While similar products on the market, such as Squarespace, Wordpress, and Medium offer great scalability and performance (Main et al., 2025), they also are a bit too involved based on our users' identified scope of simply needing to host their portfolios and collaborate respectively. This provides us a clear context for the business implications this may provide as this project aims to focus less on the content management, and more on the display of said content to end-users. Due to the scalable nature of the chosen hardware hosting and chosen tech stack, VRTS, we ensure the website will be scalable to a larger VPS system as the project's user base grows and bandwidth limitations become more apparent, detailing a need for upgrade and further testing and analysis.

If the website were to scale and elicit a buyer of the project, I could see sites like LinkedIn integrating the technology to provide a more customizable, user-friendly, platform to show off their users' work. I could see the project being sold as a hosting service for professional

content creators via subscription or other means of monetization, such as advertising on the platform. While these market and business implications fall within the scope of the design of future iterations, it falls outside the scope of this project mainly due to monetary constraints, and scope of the current user base, which is currently 2 - myself and my partner. Future iterations of the project could easily deploy to a VPS system to accommodate the potential stakeholders mentioned though!

In conclusion, this analysis phase included gathering user stories and opinions, monitoring existing products and solutions, measuring current systems potential capacity limitations, and evaluating potential business implications to define various end-users and stakeholder requirements in building a successful development blog platform that scales past its current planned iterations. We found that the current systems will support the requirements mentioned, but also acknowledged that if stakeholder interest grows and expands, a larger scale deployment to a VPS system would be needed to handle future bandwidth limitations, stakeholder requirements, and business implications. We addressed how each of the technologies used in our stack, VRTS, will aid in delivering a system that simply, securely, and efficiently meets users professional needs.

In our next iteration, we will experiment and collaborate on designs, wireframes, and schema to figure out further requirements and create an implementable design for our development phase to deliver the content.

References

3rd Wave Media. (2024, September 13). *DevCard - Bootstrap 5 vcard & portfolio website template for software developers*. UX Centred Bootstrap Templates & Themes For Developers. https://themes.3rdwavemedia.com/bootstrap-templates/resume/devcard-bootstrap-5-vcard-portfolio-template-for-software-developers/

Austin, S. (2025, January 16). *How to Determine the Ideal Size of a Web Server for Your Website*. Marketing Scoop. https://www.marketingscoop.com/website/wordpress/how-to-determine-the-ideal-size-of-a-web-server-for-your-website/

C., V. (2022, August 25). *How much bandwidth and disk space you need for a typical blog website*. FlipFall Magazine. https://flipfall.com/average-disk-space-bandwidth-website-need-hosting-plan/

Daniel. (2023, August 27). *Free open-source blog template for developers*. DEV Community. https://dev.to/danielcgilibert/free-open-source-blog-template-for-developers-218e

James. (2024, December 24). *Server administration: Understanding server load on linux*. IT Crash Course - Essential IT Knowledge. https://itcrashcourse.com/server-administration-understanding-server-load-on-linux/

Main, K., & Aditham, K. (2025, February 4). *Best blogging platforms*. Forbes. https://www.forbes.com/advisor/business/software/best-blogging-platforms/

Mirchevski, B. (n.d.). *How to determine the correct size and type of a web server*. HostAdvice. https://hostadvice.com/how-to/web-hosting/how-to-determine-the-correct-size-and-type-of-a-web-server/

Sanderson, T., & Family Member, R. (2025, February 13). Professional Blog Requirements? personal.

Sanderson, T., & Partner, R. (2025, February 13). Prefessional Blog Requirements? personal.

Sanderson, T., & Superior, R. (2025, February 13). Professional Blog Requirements? personal.

Strapi. (n.d.). Strapi CMS for content teams. https://strapi.io/for-content-teams

TechHut. (n.d.). *TechHut.tv*. https://techhut.tv/